

Course Name: Mathematical Foundation for Computer Science

Course Code: **Internal Marks: 25** **External Marks: 75** **Credit: 4**

UNIT-I

Matrix Theory :- Review of fundamentals, equivalent matrices, elementary row (column) operations, rank of a matrix by reducing it to the normal form, rank of a matrix by reducing it to echelon form.

UNIT-II

Mathematical Logic :- Connectives, Negation, Conjunction, Disjunction, conditional, bi-conditional, statement formulas, Tautology and contradiction, Equivalence formulae Normal forms: Principle conjunctive and disjunctive normal forms, Theory of inferences for statement calculus validating using truth tables.

UNIT-III

Graph Theory:- Definition of a Graph, Finite and infinite Graphs, Incidence and Degree of a vertex, Null Graph, Sub graphs, Walks, Paths, Circuits, Connected, Disconnected graphs and Components, Euler Graph, Hamiltonian Path and Hamiltonian Circuits.

UNIT-IV

Trees and Matrix Representation:- Properties of Trees, Distance and Centres in a Tree, Rooted and Binary Trees, Spanning Trees and Fundamental Circuits. Cutset, properties of a Cutset. Matrix Representation of graphs: Incidence matrix, Circuit matrix, Fundamental Circuit matrix, Cutset matrix, Path matrix, Adjacency matrix.

UNIT-V

Directed Graphs:- Definition, Some types of Digraphs, Digraphs and Binary relations, Directed paths and Connectedness, Euler Digraphs, Trees with directed edges, Fundamental Circuits in Digraphs, Adjacency Matrix of a Digraph